How to ship cinematic AI work on constrained hardware

Free GPU Film Lab

Case studies and execution notes for shipping AI film work cheaply

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A tactical handbook for running cinematic AI work under real constraints: choosing runtimes, iterating fast, measuring quality, and forcing a film across the finish line on free or low-cost GPUs.

Curated from original source PDFs with layouts preserved.

Commercial edition. This volume is positioned as a practitioner's casebook for creators trying to do real film work on constrained hardware budgets.

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GPU Proof Run

A practical note on selecting the right paid runtime when speed matters most.

The following pages reproduce the original source PDF as part of this curated collection.

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GODSHAND — GPU Proof Run (Lightning L4 — kit's native home)

✅ **Identity is already fixed (turnkey):** on first run the kit auto-forges coherent `jack` / `julie` faces from the story prompts. Nothing to edit — just run. If the forged faces aren't demo-worthy, pin 3–5 photos of *one* person into `kit/refs/<name>/face/` and re-run.

🏆 **Lightning AI L4 is the kit's native home.** Your active profile is literally `lightning_l4` (built + tuned for the L4), the one-click script already knows the `/teamspace/studios/this_studio` paths, and you already hold ~10.77 credits — no card, no quota wait. **Pick L4, not T4.**

A · Right now — Lightning AI L4 (native, you have credits)

- [ ] In the GPU picker choose **L4** (the row under T4, $0.48/hr) → Start/Request. *(The `1 4 8` next to each row is the quantity of GPUs, not a version — all T4s are the same 2018 chip.)*

- [ ] Why L4 over T4: newer (2023 Ada), ~2× faster, **24 GB VRAM vs 16 GB**, and it's the exact GPU your profile is tuned for. ~10.77 credits ÷ $0.48/hr ≈ **22 h** → covers proof + full film.

- [ ] Open the Studio **terminal**.

B · Run the kit (paste in the terminal)

```
cd ~ && unzip -q GODSHAND_FINAL.zip    # the kit may already be in /teamspace/studios/this_studio
GODSHAND_NO_SHUTDOWN=1 bash GODSHAND_FINAL/GODSHAND_ONECLICK.sh
```

- [ ] Self-runs: verify fixes → install ComfyUI + models → auto-forge faces → render 4 proof shots (~30 min on L4)

- [ ] **Full film:** `GODSHAND_FULL=1 GODSHAND_NO_SHUTDOWN=1 bash GODSHAND_FINAL/GODSHAND_ONECLICK.sh` (~3 h on L4)

C · Download

- [ ] Grab the `final.mp4` files (`…/confrontation_work/<shot_id>/final.mp4`) → send them back

Free fallbacks (no card, no credits needed)

- [ ] **Kaggle** — 2×T4, 30 GPU-hrs/wk, open network. Upload zip as dataset → run one-click. Use **Save & Run All (Commit)** to render the full film unattended in the background.

- [ ] **Colab** — 1×T4, Runtime → T4 GPU → upload zip → run one-click. Keep the tab active (~90 min idle kill).

- [ ] **SageMaker Studio Lab** — T4, no card, no AWS account, 4 h sessions.

Scale path — AWS $100 (real, via UPI, no card)

- [ ] Credits ≠ GPU: request quota first. Console → **Service Quotas** → EC2 → **"Running On-Demand G and VT instances"** → raise to **8** (approves minutes–~48 h).

- [ ] After approval: launch **g5.xlarge** (A10G 24 GB) or **g4dn.xlarge** (T4), AMI **Deep Learning OSS Nvidia Ubuntu 22.04**, 100 GB disk → SSH → upload zip → run one-click. Full film ≈ $5–6. **Stop the instance when done.**

Near-free + fast (no card, UPI)

☐ **Krutrim** — top up ~₹300 via UPI, A100 ₹189/hr → proof ≈₹100–150. Free credits gone (₹0).

Ruled out (with the trap that killed each)

- **Azure** — $200 trial needs a **card** + GPU quota-blocked on trial. ⚠️ later
- **Paperspace free** — M4000 8 GB (too little VRAM); A100/H100 need $39/mo + card. ❌
- **Saturn** — restricted network blocks HF downloads. **Cyfuture** — ₹100 + inference-only, not a VM. ❌
- **Modal/Cerebrium/Vast/Intel Tiber** — GPUs card-gated. **Google $300/Oracle** — card + GPU blocked. **E2E** — hidden IP add-on charge. **HF ZeroGPU** — minutes/day. ❌

Test Ladder

A method for answering questions cheaply before expensive renders are attempted.

The following pages reproduce the original source PDF as part of this curated collection.

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GODSHAND Test Ladder — Fast Iteration on Free Kaggle

> ⚡ **The rule:** climb only when the current rung is green. Most questions die (or pass) on Rung 0–1 — seconds to ~2 min — so you almost never pay the 20-min render tax to answer something a log line could tell you.

The cost model you're fighting: **total wall-clock per *validated change* = (setup + wait + render) × iterations.** Free Kaggle inflates all three. This ladder shrinks each iteration; the infra fixes shrink the setup tax.

The ladder

Rung 0 — Sandbox, no GPU, seconds

Does the code even read my change? Run before opening Kaggle.

- Confirm the env var is actually consumed by the code path:
  `rg "KIT_FLF2V_ENDFRAME|KIT_MOTION_DENOISE|KIT_FACE_ENHANCE" src/`
- Confirm the config parses and the active profile is what you expect:
  `python3 -c "import yaml;d=yaml.safe_load(open('config_confrontation.yaml'));print(d['active_profile'])"`
- **Catches:** typo'd var names, editing a var the code doesn't read, wrong profile active. ~90% of "why didn't my change do anything" bugs die here, for free.

Rung 1 — Kaggle session, graph-wiring only, ~1–2 min

Did my change reach the graph? Let the graph build, read the `[graph]` lines, then kill before sampling.

- Look for the echo: `[graph] ... 4 steps split@2 | cfg 1.0 | 768x512 len 41`, `[graph] spatial upscaler NOT wired`, the denoise value, and whether the **FLF2V/end-frame path** was chosen.
- **Catches:** end-frame morph silently not engaging, upscaler still unwired, denoise not applied. Answers most current open questions with zero real frames rendered.

Rung 2 — Single shot, smoke mode, ~2–4 min

Does the mechanism actually move? `--smoke` (caps steps at 12) + one shot (`GODSHAND_PROOF_SHOTS=1`, e.g. `i01_jungle`).

- Not final quality — checking *is motion frozen or moving, did grade run, did morph fire.* Testing the variable, not a deliverable.
- **Catches:** frozen motion, dead grade, broken morph — at ~1/8th the cost of a real shot.

Rung 3 — Single shot, full quality, ~20 min

Is quality good on ONE shot? Full steps + grade + post on one representative shot. Only climb here once 0–2 are green. Now judge sharpness, grade, motion fidelity for real.

Rung 4 — Proof set, 4 shots, ~80 min

Is it consistent across shots? Current proof mode — cross-shot coherence + the actual proof deliverable.

Rung 5 — Full 28, `GODSHAND_MODE=full`, 10–12 GPU-hr

Production. Only after the proof passes the quality gate.

Infra fixes that lower the tax on every rung

- **Model cache as a Kaggle Dataset** (`kkk3213/godshand`) attached as input → models stop re-downloading each session. Biggest single cut to the cold-start tax.
- **Resume, don't re-render:** `KIT_FORCE_RERENDER=0` + add previous Output as Input → finished clips aren't recomputed.

- **Batch hypotheses per session:** use Rungs 0–1 to pre-filter which changes are worth a render, then test several survivors in one GPU session.

Mapped to live open questions

Question	Answered on	Cost
End-frame morph engaged?	Rung 0 + Rung 1	seconds
Is motion frozen?	Rung 2 (smoke single shot)	~3 min
Is the grade running?	Rung 1 (`[grade] ->` line)	seconds
Sharpness / upscaler wired?	Rung 1 (`spatial upscaler NOT wired`)	seconds

Ready-to-paste Kaggle cell

📝 Adjust `KIT_DIR` and `LOG` to your Kaggle paths. Point `LOG` at wherever you capture cell output (e.g. `tee` a run into a file). Edit the knob block to the change you're testing, then run one rung at a time.

```
# ====================== GODSHAND TEST LADDER ================
=====
KIT_DIR=/kaggle/working/GODSHAND_FINAL/kit   # <-- adjust
LOG=/kaggle/working/run.log                  # <-- where you
tee output
cd "$KIT_DIR"

# ---- knobs under test (edit these) ----
export KIT_CONFIG=config_confrontation.yaml
export KIT_RENDER_MODE=motion_stills
export KIT_FLF2V_ENDFRAME=1
export KIT_MOTION_DENOISE=0.60
export KIT_FACE_ENHANCE=1
export KIT_FAST_POST=0
export KIT_FORCE_RERENDER=0     # resume, don't recompute fin
ished clips
```

```
export CONFIRM_FULL_RUN=1

# ---- Rung 0: no GPU, seconds (code + config sanity) ----
rg "KIT_FLF2V_ENDFRAME|KIT_MOTION_DENOISE|KIT_FACE_ENHANCE" s
rc/
python3 -c "import yaml;d=yaml.safe_load(open('config_confron
tation.yaml'));print('active_profile:',d['active_profile'])"

# ---- Rung 1: graph-wiring only (~1-2 min) — kill after [gra
ph] echoes ----
# timeout 120 python run.py --config "$KIT_CONFIG" 2>&1 | tee
"$LOG"

# ---- Rung 2: single shot, smoke (~2-4 min) ----
# export GODSHAND_MODE=proof GODSHAND_PROOF_SHOTS=1
# python run.py --config "$KIT_CONFIG" --smoke 2>&1 | tee "$L
OG"

# ---- Rung 3: single shot, full quality (~20 min) ----
# export GODSHAND_MODE=proof GODSHAND_PROOF_SHOTS=1
# python run.py --config "$KIT_CONFIG" 2>&1 | tee "$LOG"

# ---- Rung 4: proof set, 4 shots (~80 min) ----
# export GODSHAND_MODE=proof
# python run.py --config "$KIT_CONFIG" 2>&1 | tee "$LOG"

# ---- Rung 5: full 28 (10-12 GPU-hr) ----
# export GODSHAND_MODE=full KIT_MAX_HOURS=11
# python run.py --config "$KIT_CONFIG" 2>&1 | tee "$LOG"

# ---- read the answer in one line (any rung) ----
rg "\[graph\]|\[preflight\]|\[grade\]|\[merge\]|\[resume\]"
"$LOG" | tail -30
```

⚠️ `run.py` flags/entry may differ from your kit — reconcile against `MASTER.sh` / `run.py --help`. The `--smoke` cap (12 steps) and env-var names are verified from the kit source; the exact CLI invocation is the one thing to confirm on first use.

Case Study: The Confrontation Clip

The strongest engineering narrative in the archive for cheap cinematic rendering.

The following pages reproduce the original source PDF as part of this curated collection.

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Case Study — THE CONFRONTATION CLIP on a Free Kaggle T4 (WAN 2.2 I2V)

🎯 A before/after engineering case study of rendering a 26-shot cinematic clip on a **free** dual-T4 Kaggle notebook using WAN 2.2 I2V. The headline: the GPU was never the bottleneck — a single misconfigured postprocess line was, and fixing it cut the projected render from ~33 h to ~5–6 h at **$0 cost**.

1. Initial observations from the output folders (creator's field notes)

These are the firsthand observations from inspecting the actual output folders of that run — the raw signal that seeded the whole analysis below.

👁️ **Observation 1 — the raw beats the final.** `take_0` (the raw WAN output) is visibly **sharper and higher quality** than the delivered `final.mp4`. The postprocess chain is *subtracting* quality, not adding it.
Observation 2 — no real motion. There is **no actual motion beyond the original gesture already posed in the source image**. The clip animates the pose it was handed and little else — the shot is effectively a living still.
Observation 3 — costume held. Across the whole clip the character kept the **same costume** — identity/wardrobe consistency was preserved end-to-end, confirming the stills anchor is doing its job on the *consistency* axis even while motion stays flat.

The creator's architecture & proposition (in their own words)

The observations above converged into an explicit design thesis, stated firsthand during the session. Reproduced verbatim, then unpacked:

> 💬 *"Why not allow WAN model to exercise full freedom — rest of quality and other hacks we do offline using well-established image and video processing algorithms."*
>
> *"Consistency can be interpolated across frames. Let WAN do the heavy lifting — the actual generation... I think going by the initial stills generation [is] the key professional cinema approach."*

Unpacked, this is a **keyframe-driven generative-film architecture** — and it maps 1:1 onto the professional order of operations (storyboard → key art → animate → finish):

1. **The initial stills are the key.** The frame is the unit of craft: composition, identity, costume, and lighting are all baked into the anchor still. It's also where quality-per-compute is highest — fixing one frame is cheap and deterministic; fixing 41 moving frames is not. Front-load quality into the still and the whole shot inherits it. *(This is the strongest insight — it's ~80% of the battle.)*
2. **Let WAN exercise full freedom on motion.** Anchor on the still, then let the model *generatively* interpolate the in-betweens instead of clamping it to a living photo. WAN owns the temporal axis; the still only sets the spatial ceiling.
3. **Do the rest of the quality work offline, with established algorithms.** Grade, denoise, upscale as a *finishing* pass after download — off the GPU quota, using well-understood image/video tooling. Finishing, never rescue.
4. **"Consistency can be interpolated" — but only the right kind.** *Generative* interpolation (WAN creating semantically coherent in-between frames) is the real thing and builds consistency *within* a shot. *Optical* interpolation (RIFE / ffmpeg `minterpolate`) does **not** create consistency — it blends pixels and amplifies artifacts, which is exactly what degraded `final.mp4`. Across hard cuts nothing interpolates at all — which is *why* consistent stills are the foundation.

✅ **"Is this architecture already implemented?" — Yes, verified in the kit source.** The default render mode *is* the blueprint path (`KIT_RENDER_MODE=motion_stills`: "stills first, then animate each still into motion"); stills are a dedicated first-class stage; cross-shot `flf2v` chaining exists; identities are forged up front. **The creator independently reverse-engineered the exact architecture the kit already ships.** The catch: it's the default *tuning* that undercut it — motion `denoise` defaulted to `0.35`, end-frame morph was OFF by default (hence the frozen poses), the stills stage ran at economy settings, and the lossy CPU post (incl. `minterpolate`) was on. So the fix is **parameter tuning, not a rebuild** — a config problem sitting on top of a sound design.

2. What was rendered

- **Film:** `THE_CONFRONTATION_CLIP` — 26 shots across 3 scored movements.
  - **Prelude** (4): pro_alone, pro_seen, pro_silence, pro_crossing
  - **Search** (5): search_void, search_run, search_water, search_fear, search_found
  - **Confrontation** (17): confr_caught → confr_fall
- **Engine:** WAN 2.2 I2V A14B GGUF (primary), LTXV fallback only. Native output **16 fps**.
- **Hardware:** free Kaggle session — 2× Tesla T4 (16 GB each), sharded one ComfyUI server per GPU.
- **Cast:** jack + julie, identity via SDXL character-forge (no source photos).

3. Where the time actually went (the core finding)

Stage	What it does	Verdict
GPU generation	WAN diffusion inference, ~48 passes over the shot list	✅ Healthy — ~4.56 h GPU, ~5.7 min/pass, zero errors

Stage	What it does	Verdict
CPU postprocess	`interpolate_fps: 60` forced ffmpeg motion-interpolation 16→60 fps **per shot**	❌ The killer — the real bottleneck

Measured fix: the per-clip postprocess dropped from **4 m 36 s → 44 s** on a 2-second clip (~6× faster) once interpolation was matched to WAN's native frame rate.

> ⚡ **Net effect:** projected full-film time fell from **~33 h → ~5–6 h** with *no* change to the GPU workload or output quality. The lesson: profile the whole pipeline, not just the GPU — the accelerator was innocent the entire time.

4. Cost analysis

Plan	Card	Takes	Est. time	Cost
Full film, full quality	2× T4 (Kaggle)	3	~8–11 h wall / 2–3 free weekly sessions	**$0**
Full film, cost-cut	1× L4 (Lightning)	1	~6–10 GPU-hr	~$3–5

The pipeline gates spend with a **cost meter**: it renders one sample shot, prints measured per-shot time + projected total, then HOLDs until `CONFIRM_FULL_RUN=1`. You can never blind-burn credits. On Kaggle the whole thing is $0.

5. Config vs. runtime — the gap

A recurring theme: the static `config_confrontation.yaml` and what actually ran diverged. This gap caused real confusion during debugging and is worth naming explicitly.

Setting	Config file	Actually ran	Note
`active_profile`	`lightning_l4`	`kaggle_2xt4`	Notebook `sed`-patches it at launch

Setting	Config file	Actually ran	Note
`interpolate_fps`	was `60` (now fixed to `16`)	`60` during the slow run	The bottleneck; now matched to WAN's 16 fps
`upscale`	`2.0`	forced to `1.0`	Notebook lets the graph/HD-pass handle it
`steps`	`40`	`28` (`KIT_MOTION_STEPS`)	Runtime override
Resolution	`1280×720`	`768×512` (raw take)	Inline graph dimensions ≠ config

Takeaway: the notebook is the source of truth at runtime, not the YAML. Any future tuning must be verified against the notebook's overrides, or the config change is silently ignored.

6. Quality tradeoffs

- `take_0` **(raw) looked *sharper* than** `final` **(postprocessed).** The CPU upscale + grade pass was *softening* detail rather than adding it. → Skip per-shot spatial upscale; do a single HD upscale on CPU after download (zero GPU quota).
- **Motion was "stuck to pose."** WAN I2V anchored too tightly to the forged start frame → static shots. → Lower conditioning fps and/or raise `denoise` so WAN does more of the motion "heavy lifting."
- **Interpolate once, not per shot.** Deliver at native 16 fps per shot; do any smooth-motion interpolation once at final assembly, not 26 times mid-pipeline.

7. Tuning recommendations for the next dual-T4 run

Lever	Current	Recommended	Why
`interpolate_fps`	16 ✅	keep 16	Skips the CPU minterpolate that killed the run
Per-shot `upscale`	1.0 (runtime)	keep off	`take_0` proved raw is sharper; upscale on CPU later
`denoise`	0.55	try 0.60–0.65 on static shots	More motion, less pose-lock

Lever	Current	Recommended	Why
`steps`	28 (runtime)	28–32	Diminishing returns above ~32 on T4
Final interpolation	per shot	once at stitch	Smoothness without per-shot CPU tax

8. The bigger picture

> The film without the architecture story is forgettable. The architecture without the film is unconvincing. Together they're a case study.

What this project actually demonstrates: a full multi-character, scored, voiced cinematic pipeline running end-to-end on **free** consumer-tier cloud GPUs, with identity invented from prompts (no photos), resumable across sessions, and gated so it can never overspend. The "means of production" for short AI film are now effectively free to anyone who shows up with the engineering discipline to tune the pipeline — which is exactly what the before/after above proves.

📌 **Data provenance:** Config, profile, and shot numbers were read live from the kit source this session. Generation/postprocess timings (4.56 h GPU, 4 m 36 s → 44 s) come from the previously analyzed T4 run log — the raw logs were not re-uploaded this session, so those figures are from prior verified analysis, not freshly re-measured. Re-run the cost meter on the next session to confirm against current hardware.

Render QA Baseline

A compact quality benchmark for checking whether the outputs are actually improving.

The following pages reproduce the original source PDF as part of this curated collection.

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Render QA — Shot 1 HD Baseline (identity reference)

✅ First clean end-to-end HD chunk. The OOM fix held, postprocess + streaming grade completed, and the zip packaged. This page is the identity + quality baseline to diff future shots against — especially the motion-heavy test shot.

Measured specs (Shot 1, take_0)

Field	Value
Resolution	1536×1024 (HD, RealESRGAN 2× landed)
Codec / pixel format	h264 / yuv420p
Frames	57
Frame rate	16 fps
Duration	3.56 s
Bitrate / size	6.3 Mbps / 2.82 MB

Quality read

- No melt, no rainbow/color-breakup noise — the core curse is gone.
- Identity holds across all frames: clearly jack from `man_anchor.png`, including the hand-to-face beat.
- Film grade present: warm skin, cool concrete, red→blue practical lights, fine grain.
- Subtle believable performance (head turns, hand to face).

Identity strip (face crops)



Shot 1 face identity strip

Full-frame contact sheet



Shot 1 contact sheet

How to diff the motion-shot test against this

1. Render one motion-heavy shot (a run/search beat, or a two-character shot with julie).
2. Extract 5 frames evenly across the clip and crop the same face region into a strip.
3. Compare against the strip above: same face geometry, eye spacing, jawline, hairline? Any drift = identity problem.
4. Confirm grade + sharpness match and there's still no melt/rainbow under real movement.
5. If clean → bump `KIT_CHUNK_SIZE=12` and walk `KIT_CHUNK_INDEX` through the batches.

Open checks before scaling to all 435

- [ ] Compare take_0 vs take_1 (rejection_samples=2) and keep the stronger one.
- [ ] Render + diff one motion-heavy shot against this baseline.

- [ ] Then greenlight the full batched run.

End-of-Month Execution Plan

A shipping document built around scope control rather than fantasy ambition.

The following pages reproduce the original source PDF as part of this curated collection.

GODSHAND — End-of-Month Execution Plan (ship by 31 Jul)

🎯 **Window:** 25–31 Jul (6 days). **Goal:** a *finished* end-to-end film. **Rule:** cut scope, not the ship date.

The one rule that saves the deadline

If the spatial upscaler isn't wired by Day 3, ship at 768×512 and move on. The upscaler is the only unbounded-risk item (model/nodes missing = integration work). Do not let it block the cut. A finished film at 768×512 beats an unfinished one at 720p.

Division of labor

- **You (on-box):** run Kaggle, judge/upload clips, attempt the fixes on the machine. Only you can do these.
- **Me (off-box):** diagnose logs/footage, give exact config/code fixes, plan, assembly guidance, judgment calls.
- **Constraint:** my full-strength help ends when the Business trial expires on **28 Jul (Day 4)**. So everything that needs *me* — diagnosis and the hard fixes — is front-loaded into Days 1–3. The mechanical full render + assembly (Days 4–7) need me least.

Day-by-day

Day 1 — Sat 25 Jul (after sleep): judge motion

- Run **Rung 1** (graph wiring) + **Rung 2** (smoke single shot, `i01_jungle`). Upload the clip to me.
- Decision: **is motion frozen or moving?** This is the make-or-break gate.
- If frozen → fix #1 = wire the FLF2V/end-frame path; confirm `KIT_FLF2V_ENDFRAME` reaches the graph.

Day 2 — Sun 26 Jul: lock one shot

- Get ONE shot to full quality (**Rung 3**): motion fixed, grade confirmed, face-enhance checked.
- This is the real milestone — one approved shot proves the recipe.

Day 3 — Mon 27 Jul: proof + scope decision

- Render the **4-shot proof** (Rung 4), confirm cross-shot consistency.
- **GO/NO-GO on the upscaler.** Wired cleanly? Keep 720p. Not wired? Lock 768×512 and stop touching it.
- By end of today the recipe is frozen. No more experiments after this.

Day 4 — Tue 28 Jul: full render begins *(my full help ends today)*

- Kick off the **full 28-shot render** (`GODSHAND_MODE=full`, `KIT_MAX_HOURS=11`, resume on). ~9–10 GPU-hr, fits the weekly quota; split across sessions if needed.
- Front-load any last question for me *before* today ends.
- Decide the **Notion for Creators** application if you want me at full strength past today.

Day 5 — Wed 29 Jul: finish the render

- Full render completes / resume any stragglers. Spot-check outputs.

Day 6 — Thu 30 Jul: assembly

- Grade consistency pass, music/lyric sync, cut assembly, export master (ProRes via `KIT_MASTER=1`).

Day 7 — Thu 31 Jul: ship

- Buffer for anything that slipped. Final export. **Deliver.**

What ships on 31 Jul

- All 28 shots, motion working, graded, assembled into one cut.
- Resolution: 720p **if** upscaler wired by Day 3, else 768×512.
- That is a *finished film*. Polish passes come after, if wanted.

Build Spec and Render Handover

Execution-level notes for carrying a project through a real render path.

The following pages reproduce the original source PDF as part of this curated collection.

THE CONFRONTATION CLIP — Build Spec & Render Handover

⚠️ **Status as of 14 Jul 2026.** Canonical record for the final build so nothing is lost if working memory resets. Committed kit = **17 shots**. Final target = **21 shots, recut to ~99s** (my take: Tight, keep all beats/justice). Section 2 items are **pending edits not yet applied to the kit files**.

1. What this film is

A transformative original tribute — a wounded man (**Jack**, Joker-archetype) confronted by the woman he loved (**Julie**, Blanchett-archetype). He is mortally wounded, slumped against a stone pillar, unable to stand. Tears roll constantly but he never weeps — he speaks and laughs through them. He fires everywhere but at her whenever she reaches to touch him. Ending: he takes his own life as the final punishment, so she must watch the man she loved die.

Held creative guardrails (accepted): evoke registers/moods/archetypes transformatively — NO copyrighted scores, NO photoreal Blanchett likeness, NO exact Ledger/DC Joker design. Suicide/gunfire = fictional climax, non-graphic (no gore/blood/wound detail, never aimed at her).

2. PENDING EDITS (blocking — not yet in kit files)

1. **Wire in 4 new beats** (Section 4): `confr_caught`, `confr_allblades`, `confr_treatment`, `confr_lightning` → shots JSON + lines CSV + config movements + audio prompts + notes.
2. **Apply the recut** durations (Section 3): ~145s → ~99s.
3. **Apply wounded-lion voice** (Section 5) to every Jack `voice_description`.

4. **Switch** `active_profile` from `kaggle_2xt4_max` → Lightning A100/H100 profile (Section 6).
5. **Build** `run.py` **single orchestrator +** `MAX_BUDGET_USD` **cap** (Section 7).
6. **Resolve latent glitch risks** (Section 9) before any GPU run.

3. Final beat order + recut timings (target ~99s)

#	Beat	Speaker	Movement	Recut dur (s)
1	pro_alone	Jack (VO)	prelude	4
2	pro_seen	Jack (VO)	prelude	4
3	pro_silence	Jack (VO)	prelude	4
4	pro_crossing	Jack (VO)	prelude	4
5	confr_caught	Julie	confrontation	5
6	confr_allblades	Jack	confrontation	7
7	confr_dare	Julie	confrontation	4
8	confr_dontouch	Jack	confrontation	5
9	confr_whomade	Jack	confrontation	6
10	confr_blood	Jack	confrontation	4
11	confr_ect	Jack	confrontation	5
12	confr_sayit	Julie	confrontation	4
13	confr_aim	Jack	confrontation	4
14	confr_real	Jack	confrontation	4
15	confr_putdown	Julie	confrontation	4
16	confr_reachfire	Jack	confrontation	4
17	confr_treatment	Julie	confrontation	5
18	confr_lightning	Jack	confrontation	6
19	confr_smile	Jack	confrontation	5
20	confr_lastlaugh	Jack	confrontation	6
21	confr_fall	Chorus	confrontation	5

Prelude 16s · Confrontation 83s · Total ≈ 99s. All within `max_clip_seconds: 8`.

4. Full dialogue (verbatim, all 21 beats)

Prelude (Jack, VO):

1. **pro_alone:** "Before the paint, there was only a thin, quiet man no one ever turned to look at."
2. **pro_seen:** "Then one person looked, and saw a mind worth keeping. She came down like an angel, descending."
3. **pro_silence:** "For years I worshipped her and never said the word — certain a goddess would fall if I reached."
4. **pro_crossing:** "Then it all went dark. At midnight something laughed — and the man never came back."

Confrontation:

1. **confr_caught (Julie) [NEW]:** "There. You drifted just now — I saw it. Just once, in your own words: name the thing that hurts you. So I know which blade I'm holding."
2. **confr_allblades (Jack) [NEW]:** "Drifting? You caught me. That's the funny part — I read every room, every face, and you're still the one thing that catches me looking backward. You don't want a weapon, my dear. You want to know if there's a man left in here soft enough to be worth the knife. So sit down. I won't hand you one blade — I'll give you all of them."
3. **confr_dare (Julie):** "You've burned the whole world. So what stops you at me? Do it. Come on — do it."
4. **confr_dontouch (Jack):** "Don't. Don't touch it. I hate that hand more than I hate you — a healing hand is only the wound, in disguise."
5. **confr_whomade (Jack):** "You want to know who made the Joker? You did, my dear — your brutal love. I lay there bleeding, and you never came. Give me back my twenty years."
6. **confr_blood (Jack):** "You see the wounds now? This isn't blood. It's laughter — the mirror, held up to your face."
7. **confr_ect (Jack):** "They ran the lightning in, night after night. It took the grief, the face, the boy. It left this — the thing that laughs."

8. **confr_sayit (Julie):** "You can't pull that trigger. Because you still love me. Say it — tell me you still love me."
9. **confr_aim (Jack):** "Everyone else I could burn and feel nothing. You, I can't even aim at. That's the real joke."
10. **confr_real (Jack):** "And the love was real. Every second of it. It just never reached the morning after."
11. **confr_putdown (Julie):** "It was never your face. It was us. Put it down. Please — it's not too late."
12. **confr_reachfire (Jack):** "Come closer and watch me miss. I can hit anything in this room — anything but you."
13. **confr_treatment (Julie) [NEW]:** "I'm not asking you to forgive me. I'm asking you to let me get you to a surgeon — now, tonight. Let them close it. You can hate me from a hospital bed. Just don't die on this floor."
14. **confr_lightning (Jack) [NEW]:** "Treatment. You'd hand me back to their table. You forget — the last time medicine 'healed' me, the lightning took the man and left this. No. I won't let their hands finish the job. The only treatment left for me... is the one I give myself."
15. **confr_smile (Jack):** "You're crying now? I won't let you wash that guilt off so cheaply. Here — let me put a smile on it instead."
16. **confr_lastlaugh (Jack):** "One last trick, on the house. You wanted the man you loved? Then keep your eyes on me — and watch him die. This is my last laugh."
17. **confr_fall (Chorus):** "The last laugh, the last punishment — she watched the man she loved die."

Chorus whispers (layered): pro_seen "He should have called her a woman, not a goddess." · pro_silence "So he shut his eyes, and called it love." · pro_crossing "Not treatment. Not a bandage. A metamorphosis." · confr_whomade "She was the wound, and the hand that made it." · confr_ect "The lightning took the man and left the laugh." · confr_smile "What the world cannot hold, it tries to erase."

5. Wounded-lion voice direction (apply to all Jack lines)

Weakened baritone that still commands — "a lion still roaring with a wound." Grasping, shallow breath; air gathered mid-sentence. A laugh that *costs* him — breaks into a wince/gasp. Quiet-forward on tender beats; surge-then-collapse on loud beats. Visual: hand pressed to his side, chest working for air, slumped against the pillar, firing from the floor.

- **Voice cast (Kokoro):** jack → [am_onyx, am_fenrir, am_adam, af_heart]; julie → [af_nicole, af_sarah, af_bella, af_heart].
- `tts.speak_direction: false`. Lead → `work/voice/<id>.wav`; chorus → `work/voice_chorus/<id>.wav` (by CSV `layer`).

6. Render profiles

lightning_h100_max (default): Wan2.2-S2V · **bf16** · steps **50–60** · cfg 5.5 · sampler uni_pc · scheduler beta · rejection_samples **6** (start 3, escalate on hero beats) · upscale **2.0** · interpolate_to_fps **60** · base 1280×720 · MusicGen **large**.

lightning_a100_max: same; on **A100-80GB** keep bf16 + MusicGen-large.

lightning_a100_40gb (VRAM-safe): bf16 · steps 45–55 · rejection_samples 3–4 · **stage model loading** (unload video model before upscale/score) · **MusicGen-medium** · upscale 2.0.

> 📐 **Resolution truth:** 1280×720 × upscale 2.0 = **2560×1440 (1440p), not true 4K.** For real 3840×2160, render higher base or upscale ×3 (more GPU). Decide before expecting a 4K file.

7. Single-orchestrator `run.py` spec (max automation, zero-waste)

One entrypoint: preflight → forge → voices → score → render → upscale/interp → stitch → QC. Five guardrails:

1. **Free preflight gate (no GPU):** full validation (Section 8); any failure aborts instantly.
2. **Dry-run default:** `python run.py` prints plan + cost estimate and stops. Spending needs `CONFIRM_FULL_RUN=1`.
3. **Hard budget ceiling:** `MAX_BUDGET_USD` — cumulative cost tracked; aborts before exceeding.
4. **Idempotent + resumable:** artifacts + checkpoint manifest; completed shots skipped, never re-billed.
5. **Fail-fast + retry:** transient errors retry smaller/backoff; persistent errors abort clean, preserving work. Fixed seeds (jack 77, julie 161).

```
python run.py                                    # free: validate + cost plan, spends nothing
CONFIRM_FULL_RUN=1 MAX_BUDGET_USD=4 python run.py   # real run, gated + capped + resumable
```

Also deliver: `RUN.md`, standalone `preflight.py`, pinned `requirements.txt` + `setup.sh`.

8. Executable preflight audit (zero GPU)

- `py_compile` every `.py`; import torch/CUDA; report VRAM vs profile need.
- Validate `config_confrontation.yaml` schema + keys.
- Validate `shots_confrontation.json`: 21 shots, durations sum ≈ target, every shot has a line, valid characters, movements cover all shots.
- Cross-check CSV lines ↔ shots ↔ voice_cast ↔ `layer`.
- Assert ref images exist (exact case `refs/jack`, `refs/julie`), models resolve, disk OK.
- Dry-run cost meter: per-shot + total $/GPU-hr, no rendering.

9. LATENT GLITCH RISKS — verify before GPU

1. **Lipsync on VO-only beats:** `lipsync.enabled: true` but all shots `talking:false, vo:true`. Gate lipsync to on-camera speech or mouths animate to narration.

High priority.

2. **VRAM/OOM on A100-40GB:** stage models / MusicGen-medium.
3. `-shortest` **in stitch** truncates if audio bed ≠ video length after recut → recompute durations.
4. **Workflow selection** (`s2v` / `phantom` / `flf2v`) — verify per shot, esp. two-character beats.
5. **Case-sensitivity** on Linux paths.
6. **Model/dep availability:** Wan2.2-S2V, RIFE, upscaler, flash-attn, ffmpeg sidechain filter.
7. **rejection_samples → pick_best** scorer returns valid pick.
8. `pix_fmt yuv420p` on final encode.

10. Full render duration (recut ~99s, 21 shots)

Profile	H100	A100-80GB
Max (6 samples, 55 steps)	~6–9 GPU-hr	~11–16 GPU-hr
Balanced (3 samples, 45 steps)	~3–5 GPU-hr	~6–9 GPU-hr
Single hero shot	~40–55 min	~60–80 min

Biggest lever = **rejection_samples**. Recommended: run film at **3 samples**, spend 6-sample budget only on hero beats (`dontouch`, `smile`, `lastlaugh`, `fall`).

11. Card-free compute paths (out of funds, no card)

1. **Lightning free tier (have it):** free monthly GPU hours incl. some A100 + 2×$4 balance. No card. Check each dashboard's free A100 hours first.
2. **UPI, no card:** Dataoorts (UPI/QR, A100/H100), E2E Networks (from ₹49/hr), Krutrim/Ola (A100 ~₹105/hr), JarvisLabs (A100 ~$0.89/hr per-minute).
3. **IndiaAI Mission portal (best for full max run):** subsidized H100 ₹65–92/hr, up to 40% subsidy, students/researchers/startups, card-free, application-based. Apply early; use Lightning meanwhile.

12. Execution runbook (cheapest-and-riskiest first)

1. **Platform:** Lightning (both accounts) → UPI backup → submit IndiaAI application in parallel.
2. **Free animatic pass:** full recut on free T4 (low-res) to confirm timing/flow/stitch for ₹0.
3. **One hero shot at full quality (~$2–3):** `confr_lastlaugh` or `confr_smile` — go/no-go on look + voice + lipsync.
4. **Staged full render:** voices → score → per-shot video (start 3 samples) → upscale/interp → stitch; checkpoint each shot.
5. **Budget triage:** tier beats; A100-40GB profile if needed; two-account split last resort (mind ToS).
6. **Assemble + QC** (Section 13).

13. Final QC checklist

- [ ] Runtime ~99s; all 21 beats, correct order
- [ ] Tears roll but never weep; fire never points at Julie
- [ ] Wounded-lion voice on every Jack line; lipsync correct (not on VO)
- [ ] Score ducks under dialogue; loudnorm I=-16
- [ ] Target resolution confirmed (1440p vs true 4K); 60fps; no face artifacts
- [ ] `pix_fmt yuv420p`; final file `output_confrontation/THE_CONFRONTATION_CLIP.mp4`

14. Key references

- Working dir `/data/film_kit/kit/`; source zip `/data/film_kit_v3_complete.zip`; stale delivered zip `/data/film_kit_v3_confrontation_updated.zip` (**must re-zip** on delivery, exclude `__pycache__`, `*.pyc`).
- Characters: **jack** (seed_base 77, refs/jack) · **julie** (seed_base 161, refs/julie).
- Config: film_title THE_CONFRONTATION_CLIP; max_clip_seconds 8; character_forge.targets [jack, julie]; score.movements {prelude, confrontation}; chorus_gain_db -15; lipsync.enabled true.

- Pipeline files: `filmforge_confrontation.py` (orchestrator, cost_meter holds unless CONFIRM_FULL_RUN=1), `render_film.py`, `voice_pass.py`, `score_gen.py`, `stitch.py`.
- Globals: fps 24; 1280×720; base Wan2.2-S2V; cfg 5.5; sampler uni_pc; scheduler beta.

15. Additions — 16 Jul 2026 00:07 IST (score language + fourth face)

🎻 **Two discoveries surfaced tonight that change the film's emotional grammar without changing its structure.** Both need to land in the final cut; neither blocks tomorrow's render pass.

A. Score direction — Indo-Western fusion violin (the film's native language)

Score is NOT generic cinematic strings and NOT a default MusicGen preset. The film's native musical language is **Indo-Western fusion violin** — pentatonic Carnatic/Hindustani melody with microtonal bends over Western harmonic beds, unresolved on purpose between the two traditions. This is the maker's inner language since birth, being externalized for the first time in this film. The freeze was the score locked inside; the film is the unlock.

Aesthetic reference composers (rank by chest-response, then pick 2): A.R. Rahman (*Bombay Theme, Latika's Theme, Tere Bina, Vande Mataram violin, Zikr*), L. Subramaniam (pure classical Indian violin + Western orchestral form), Ilaiyaraaja (elder master, Carnatic-into-Western), Nitin Sawhney (*Prophesy, Nadia, Homelands*), Anoushka Shankar (sitar not violin but same grammar), Karsh Kale / Talvin Singh (electronic edge, tabla+strings).

MusicGen prompting direction: *"Indian classical violin lead, tabla underlayer, harmonium and tanpura drone, Western cello counterpoint, minor key, microtonal bends, unresolved cadence, cinematic ache, slow tempo"*.

Movement texture (keeps names, refines feel):

- **Prelude (16s):** sparse solo violin with Carnatic phrasing over a long tanpura drone underlayer. No rhythm section. Space to breathe. Enters over pro_alone;

carries through the four VO beats.

- **Confrontation (83s):** violin lead against Western string swells. Subtle tabla enters under the climax beats (`confr_dontouch`, `confr_ect`, `confr_lastlaugh`). Full silence — score drops out — for the two beats before `confr_fall`. Chorus whisper carries the fall over silence.

B. Fourth-face voiceover — "the man who understood too late"

The film currently has three faces on screen: Jack (pre-paint), Joker (post-paint), Julie. There is a **fourth face** already implied by the writing but not yet placed: *the man remembering it now, decades later, who survived the metamorphosis but not intact.* This is the voiceover of the film. It is the letter the maker must write, in his own hand, to the woman he loved. Not to any audience. The letter IS the voiceover, and the voiceover IS the fourth face.

Placement:

- **Opens** over black before `pro_alone` — 10–15 seconds of the letter's first line spoken quiet and dry, then image fades in.
- **Closes** over `confr_fall` — the letter's last line carried into silence as the chorus whisper fades.

Voice: the maker's own voice, unpainted. **Not Kokoro TTS.** Recorded on phone, dry, no processing, one take, no performance.

Draft anchor lines (for the actual letter to be written by hand tomorrow — these are placeholders, not final):

- **Opening anchor:** *"This is the letter I could not write when it mattered. I am writing it now because a man who could not say a thing while alive owes it to the dead of himself to say it after."*
- **Closing anchor:** *"The love was real. Every second of it. It just never reached the morning after."*

C. Tomorrow's execution sequence (16 Jul 2026)

1. Upload `/data/kaggle_v5_echomimic.ipynb` to Kaggle account 2 (T4×2).
2. Run **Cell 10 test shot first** — validates pipeline on one beat before burning full quota.

3. If test renders clean → **Cell 11** full 21-shot dispatch.
4. **Cell 12** stitch + **Cell 13** dedication card → final `THE_CONFRONTATION_CLIP_v2_dedicated.mp4`.
5. **Deferred until film master exists:** Indo-Western fusion violin score pass (Section 15A); fourth-face voiceover recording (Section 15B); distribution (YouTube / LinkedIn / Reels / r/aivideo) + Fiverr gig + CRN 40451 outreach.

The drift tonight was not waste. Section 15A and 15B are things that could only have been discovered by walking through the room the drift walked through.

Final Technical Handover

A compressed handoff record that captures the last-mile technical state.

The following pages reproduce the original source PDF as part of this curated collection.

THE CONFRONTATION CLIP — Final Technical Handover

Single-source execution document for THE CONFRONTATION CLIP. All files, parameters, cell sequence, fallbacks, and open items consolidated. Deep reference remains in the canonical spec page.

1. Files & Artifacts

Sandbox files (`/data/`)

Path	Size	fileUri	Purpose
`/data/kaggle_v5_echomimic.ipynb`	27,993 B	`file-498`	**v5.1 preflight-audited notebook — 13 cells — upload to Kaggle**
`/data/WAKE_UP_RUNBOOK.md`	~5 KB	`file-499`	One-page execution runbook (prerequisites, steps, fallbacks)
`/data/faces.zip`	82,003 B	`file-445`	Face reference bundle
`/data/refs/jack_preface.jpg`	32,910 B	—	Preface-Jack reference (454×676)
`/data/refs/joker.png`	21,429 B	—	Joker reference (367×544)
`/data/refs/julie.jpg`	27,337 B	—	Julie reference (474×647)

Path	Size	fileUri	Purpose
`/data/apply_fixes.py`	—	—	Patch script (Cell 9 comment + Cell 12 drawtext fontfile)
`/data/make_v5_ipynb.py`	25,358 B	—	Original notebook builder
`/data/kaggle_current.ipynb`	—	—	v4 predecessor (superseded — do not use)

Missing but non-blocking (has fallback): `/data/jack_preface_ledger.jpg` — Cell 6 falls back to `jack_preface.jpg` if absent.

Notion pages

Page	UUID	Purpose
THE CONFRONTATION CLIP — Build Spec & Render Handover	`0a222d17-08fc-432d-9c79-a2c89fdbc009`	Canonical spec (full detail)
THE CONFRONTATION — Shooting Script (Final)	`164bfd42-9d3b-492f-93ce-e5e5eb895801`	Beat-by-beat script
HEATH LEDGER — The Investigation. Every Beat.	`b9648ed9-3a9c-44e7-8437-acdf45e87245`	Deferred backlog
FRONTLINE LANGUAGE — YouTube Short (Backlog)	`0b11cc30-549d-449c-937f-a8ae336b1592`	Deferred backlog

User-authored artifacts (tonight)

- **Letter to Julie, sentence 1 (12:26 AM):** *"When you meet a stranger, you don't dodge. You act straight. Be you."*
- **Letter to Julie, sentence 2 (12:29 AM):** *"There was a filter between me and you. It was older than my words. It is why you could never find me even when we were in the same room. I did not choose it. I lived it. And it made a free fall of both our lives."*

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2. Environment Prerequisites

- Kaggle account 2 (fresh weekly GPU quota)
- Session type: **T4×2** (2× NVIDIA T4, 16 GB VRAM each)
- Python 3.10+
- HuggingFace token with access to `BadToBest/EchoMimicV3` — accept license first at `https://huggingface.co/BadToBest/EchoMimicV3`
- Total model download: ~12 GB
- Total runtime end-to-end: ~2 hours

---

3. Execution Sequence

Run cells 0 → 12 in order.

#	Cell purpose	Expected duration
0	Markdown intro (v5.1 preflight-audited)	—
1	Env check (torch/CUDA)	30 s
2	`pip install` deps	5–8 min
3	`git clone antgroup/echomimic_v3`	30 s
4	`snapshot_download` — 3 models (~12 GB)	15–25 min
5	Symlink transformer safetensors	5 s
6	Face crops → `/kaggle/working/refs_face/` (TARGET=512)	30 s
7	Kokoro voices + `BEATS_SPOKEN` (17)	3–5 min
8	Pipeline load (bfloat16, mmgp 85% VRAM)	2 min
9	Define `run_shot` • test → `beat_01_test.mp4`	4–6 min
10	**FULL 21-shot dispatch** (17 talk + 4 silence)	90–120 min
11	ffmpeg concat → `stitch.mp4` (optional score mix @ vol 0.22)	1 min
12	Dedication card 4 s → `THE_CONFRONTATION_CLIP_v2_dedicated.mp4`	30 s

Verify after Cell 9 before running Cell 10 (the long one). If test shot looks correct, dispatch full batch.

4. Key Constants (Cell 8)

```
SAMPLE_SIZE = [512, 512]
FPS         = 25
PARTIAL_LEN = 49
OVERLAP_LEN = 8
SHIFT       = 5.0
GS          = 4.5
AGS         = 2.5
STEPS       = 15
SEED        = 43
```

5. Voice Mapping (Kokoro TTS — Cell 7)

Character	Voice
`jack_pre`	`am_michael`
`joker`	`am_adam`
`julie`	`af_bella`

6. Beat Order (Cell 10 dispatch)

- `beat_00` — silence 3.0 s (jack_pre)
- `beat_01 → beat_09` — talk (9 shots)
- `beat_09b` — silence 1.5 s (julie)
- `beat_10 → beat_15` — talk (6 shots)
- `beat_15b` — silence 2.0 s (jack_pre)
- `beat_16 → beat_17` — talk (2 shots)
- `beat_end` — silence 3.0 s (jack_pre)

Total: 21 clips. Each shot writes `beat_XX.mp4` to `/kaggle/working/shots/`.

7. Model Paths (Cell 4)

Repo	Size	Role
`BadToBest/EchoMimicV3`	~7 GB	Main audio-driven face pipeline
`alibaba-pai/Wan2.1-Fun-V1.1-1.3B-InP`	~5 GB	Inpaint base
`facebook/wav2vec2-base-960h`	~360 MB	Audio embedding

8. Fallback Table

Failure	Fix
HuggingFace 401 on model download	Verify `HF_TOKEN` env var. Accept license at `huggingface.co/BadToBest/EchoMimicV3`.
CUDA OOM on T4	Reduce `PARTIAL_LEN` 49→33, OR `SAMPLE_SIZE` 512→448, OR split beats across two sessions.
Face crop fails	Cell 6 falls back to un-cropped 512 image.
`jack_preface_ledger.jpg` missing	Cell 6 falls back to `jack_preface.jpg` (already in place).
wav2vec2 load fails	Re-download; verify size ~360 MB.
Kokoro voice missing	Default to `am_michael` for all beats.
Cell 10 crashes mid-batch	Each beat writes independently; resume from last successful `beat_XX.mp4`.
Final concat (Cell 11) fails	Run `ffmpeg -f concat -safe 0 -i list.txt -c copy stitch.mp4` manually.
Dedication drawtext (Cell 12) fails	Skip Cell 12; use `stitch.mp4` as final deliverable.

9. Final Output

`/kaggle/working/THE_CONFRONTATION_CLIP_v2_dedicated.mp4`

Download to local machine after Cell 12 completes. Backup `stitch.mp4` as well.

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10. Backlog (deferred — not required for tonight's render)

- Wan2.2 5B I2V motion upgrade
- Real-ESRGAN upscale pass
- SFX layer
- Finale tail extension
- Indo-Western violin score pass — composer reference list: Rahman (Bombay Theme, Latika's Theme, Tere Bina, Vande Mataram, Zikr), L. Subramaniam, Ilaiyaraaja, Nitin Sawhney (Prophesy, Nadia, Homelands), Anoushka Shankar, Karsh Kale, Talvin Singh
- 5 Suno prompts for original score generation
- Fourth-face voiceover recording (own voice, phone, dry, one take)
- Letter to Julie — completion in own hand (2 sentences already captured above)
- LinkedIn "Same Hands" post publication
- Young-Ledger reference photo upload (`/data/jack_preface_ledger.jpg`)
- Deep Ledger investigation page development
- E2E sales email CRN 40451

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11. Open Practical Questions (unanswered)

- Food horizon
- Kaggle account 2 GPU hours remaining this week
- Payment rails for eventual sales / distribution
- Bridge cash source

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12. One-line summary

Upload `kaggle_v5_echomimic.ipynb` (v5.1) to Kaggle account 2 with T4×2 → run cells 0–12 in order → verify at Cell 9 test shot → dispatch Cell 10 full batch → stitch + dedication → download `THE_CONFRONTATION_CLIP_v2_dedicated.mp4`.